Exercise sheet 2

Classroom task 2.1: What happens to the solution (x_1, x_2) of the linear system

$$Ax := \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} =: b,$$

if we add a column to the matrix on the left hand side,

$$\tilde{A}x := \begin{pmatrix} 1 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}?$$

How can we represent the solution to the new system?

Exercise 2.1: Consider the linear optimization problem

$$\max\{2x_1 + x_2 \mid 2x_1 - x_2 \leq 12,\ 0 \leq x_1 \leq 8,\ x_2 \geq 0\}.$$

Transform the problem into standard form and explicitly define c, b and A as in the lecture notes. Is the problem feasible? Is the problem bounded?

Exercise 2.2: Determine the linear program that was solved graphically in Figure 1. Standard form is not required.

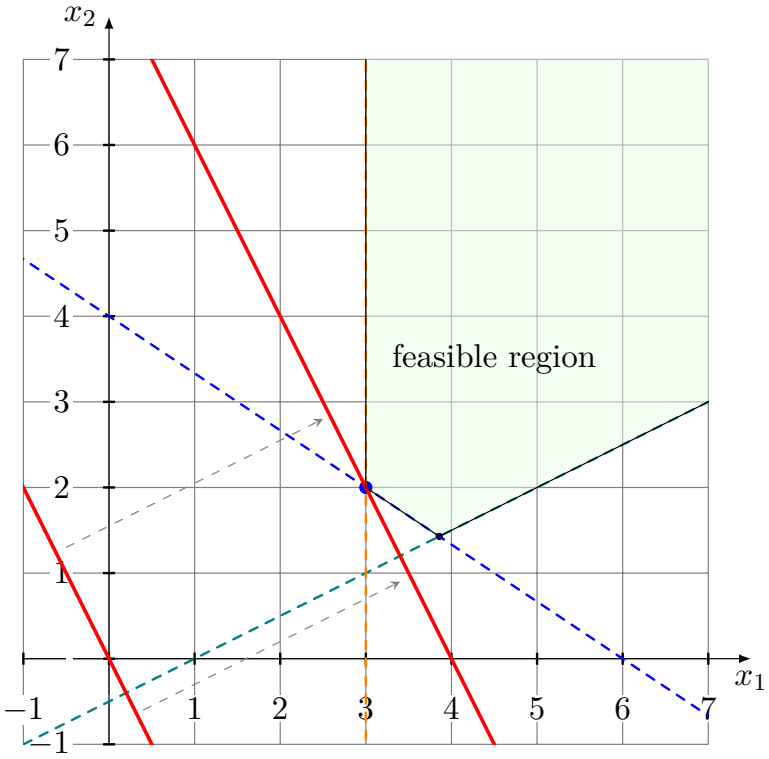


Figure 1: Graphical solution of a linear program.

Exercise 2.3: Solve the following linear program graphically:

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 5x_2 \\ \text{subject to} \quad & x_1 + 2x_2 \leq 12, \\ & 3x_1 + x_2 \leq 16, \\ & 0 \leq x_2 \leq 4, \\ & x_1 \geq 0. \end{aligned}$$

Submission deadline: 19.10.2026, 12:00